many products promising a zero-trust approach, but the core idea is still valuable.

Although implementing a zero-trust approach throughout an organization is ideal, this can’t always be relied upon and a service such as the link-preview microservice shouldn’t assume that all requests are safe. To prevent the link-preview service being abused for SSRF attacks you should validate URLs passed to the service before making a HTTP request. This validation can be done in two ways:

- You can *whitelist* URLs against a set of allowed hostnames, domain names, or (ideally) strictly match the entire URL. Only URLs that match the whitelist are allowed. This approach is the most secure but is not always feasible.
- You can *blacklist* URLs that are likely to be internal services that should be protected. This is less secure than whitelisting for several reasons. First, you may forget to blacklist some services. Second, new services may be added later without the blacklist being updated. Blacklisting should only be used when whitelisting is not an option.

For the link-preview microservice there are too many legitimate websites to have a hope of listing them all, so you’ll fall back on a form of blacklisting: extract the hostname from the URL and then check that the IP address does not resolve to a private IP address. There are several classes of IP addresses that are never valid targets for a link-preview service: